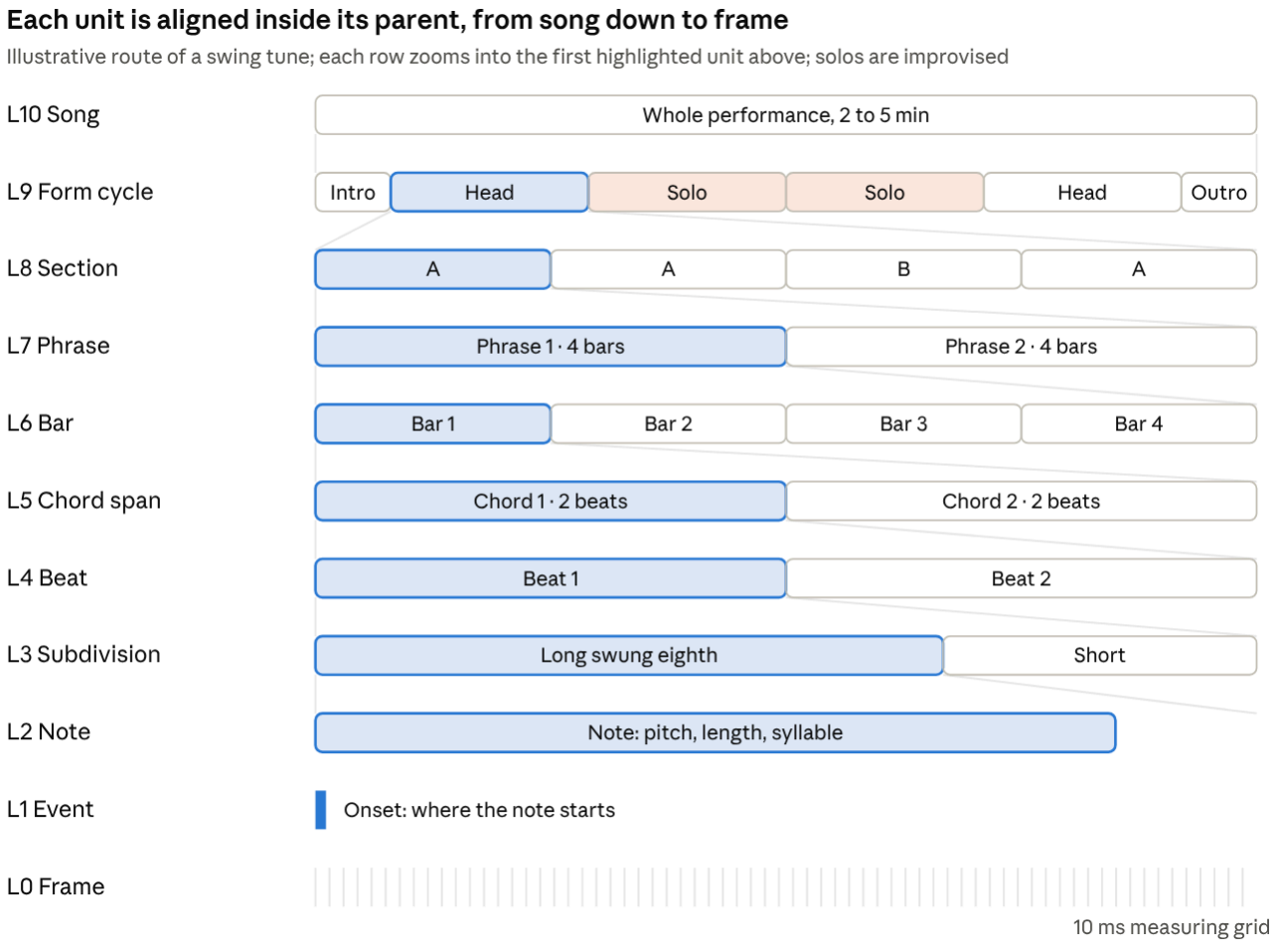
Audio Reverse Engineering — Units and Batch Alignment

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A recording is aligned to its sheet music in batches, unit by unit, never as one whole. Units nest from a 10 ms frame up to the whole song. Each unit is matched inside its parent, gets its own confidence and a mode (written or improvised), and can then be looped, searched and mixed.

The unit ladder



unit ladder · 11 levels, L0 frame to L10 song

The solos (shaded) are judged only from chord span upwards, so improvisation is never counted as a wrong note.

Units in detail

Typical lengths are for a 4/4 swing tune at 150 bpm, like *Grikiai* (one beat = 0.4 s, one bar = 1.6 s).

Level	Unit	What it is	Typical length	Boundary found by	Compared on
L0	Frame	The measuring grid, not a musical unit	10 ms	Fixed	Pitch, chroma, loudness, onset strength
L1	Event	A note start, drum hit or syllable onset	20–200 ms	Onset detection per stem	Onset time
L2	Note	One pitch with a duration (and its syllable, for voice)	0.1–2 s	Onset + offset + pitch change	Pitch, start, length, syllable
L3	Subdivision	Position in the beat: straight or swung eighths, triplets	0.13–0.2 s	Tempo map + onset positions	Swing ratio, feel
L4	Beat	The pulse; anchors the tempo map	0.4 s	Beat tracking, checked against the sheet's meter	Beat time
L5	Chord span	How long one chord lasts (harmonic rhythm)	2 beats to 2 bars	Chroma and bass changes, the sheet's chord symbols	Chord name, function
L6	Bar	One measure of the meter	1.6 s	Downbeat tracking	Bar time, meter
L7	Phrase	A melodic sentence or lyric line	2–8 bars	Rests, cadences, breaths, lyric lines	Melody contour, lyric line
L8	Section	Intro, A, B, bridge, solo, outro	8–16 bars	Repetition, harmony and texture changes; rehearsal marks	Section label, chord sequence
L9	Form cycle	One pass through the tune's form; a jazz "chorus"	32 bars for AABA	Return to the form's first chords	Form, order of sections
L10	Song	The whole performance: head, solos, head	2–5 min	File	Arrangement

Every level above L0 is a container for the level below it, so each note knows its beat, chord, bar, phrase, section and chorus.

Batch alignment

We never align a whole recording against a whole sheet in one pass. Alignment runs **top-down in batches**: each unit is matched to its counterpart on the sheet, then the units inside it are aligned within that window. A mistake stays inside the unit where it happened.

1. **Form (L9–L10).** Find which form cycles and sections the recording plays, and in what order. The recording may repeat the head, add solo choruses, cut a section or follow D.S. and coda marks differently from the sheet. The result is a *route*, such as “Intro, A A B A (head), A A B A (solo) × 2, A A B A (head), Outro”.
2. **Sections and phrases (L7–L8).** Align each section on the route to its sheet section using harmony and melody features, independently of the others.
3. **Bars and chord spans (L5–L6).** Inside each section, place downbeats and chord changes.
4. **Beats and subdivisions (L3–L4).** Inside each bar, place beats and measure the swing.
5. **Notes and events (L1–L2).** Inside each beat window, match notes and syllables, only where the unit is written (next section).

Each unit carries its own result: sheet reference (bars m–n), audio start and end, alignment confidence, mode, key, chords and local tempo. These records form a *unit index* per job, stored with the job manifest. A low-confidence unit is flagged for review without pulling its neighbours down.

Scoring by level. Metrics are computed level by level: section-order accuracy, section, phrase and bar boundary error, chord-span accuracy, beat error, and note F-measure. The scorecard shows each level separately, so it’s visible whether the app understands the form but misses notes, or the other way round.

Improvisation

In jazz swing, the sheet fixes the form and the harmony, but players are free inside it. Every unit from phrase up is therefore given a **mode**, which decides the lowest level it is judged at.

Mode	What happens	Judged down to	Not counted as error
Written	The melody is played or sung as notated	L2 notes	Timing feel within the tempo map

Mode	What happens	Judged down to	Not counted as error
Embellished	The written melody with ornaments, anticipations, rephrased rhythm	L2 notes, with wider timing tolerance and skeleton matching (main notes on strong beats)	Added grace notes, slides, delayed or anticipated entries
Improvised	A solo, scat or ad-lib over the form; notes not on the sheet	L5 chord spans (harmony fit, meter, form)	Every note that fits the harmony
Comping	Accompaniment rhythm and voicings chosen by the player	L5 chord spans and L3 feel	Voicing and rhythm choices

- **Where the mode comes from.** First from the sheet: solo sections, slash notation and chord-only bars mean improvised or comping. Otherwise it's detected: high chord agreement but low note agreement means improvised. A musician can confirm or change it.
- **What the app still delivers for improvised units.** The full performance transcription (every note it hears) without a score truth to grade it. Notes are graded only on how well they fit the chords (chord-tone ratio) and the meter.
- **Note F-measure is computed only over written and embellished units.** An improvised solo can't lower note accuracy, and it can't inflate it either.

Independent timing truth

If gold timings come from aligning the sheet to the recording, and the app is graded with a similar alignment, both make the same mistakes and timing scores look better than they are. Four rules keep the two apart.

1. **One fully timed benchmark song per language.** A musician marks every bar downbeat and every sung word by ear (tap along, then correct), with no software suggestion shown first. This is the independent reference for beat, note and word timing.
2. **Derived timings count fully only from bar level up.** Timings derived by aligning a sheet to a recording, then spot-checked, count fully for bars, phrases and sections (L6–L9). Below bar level (beats, notes, words) they count half, permanently. Only fully timed songs give full evidence there.
3. **Gold is never made with the app's aligner.** Gold derivation uses a separate, frozen score-follower with different features, and its output is corrected by the musician.

4. **Spot-checks include unannounced points.** Besides planned anchors (phrase starts, one per 4 bars), the musician checks 10 random positions per song, so drift between anchors gets caught.

The report shows timing metrics for benchmark songs and derived songs side by side. If they disagree by more than 20 ms in median, the derived timings are treated as unreliable until reviewed.

Loops and mixes

Once units are aligned with confidence, each one becomes a building block: something to loop, export, search for and recombine.

- **Loop candidates** are units from L5 to L9 (chord span, bar, phrase, section, chorus). A loop's quality depends on: start and end on a downbeat, no note sustained across the loop point (or a short crossfade where one is), matching loudness and timbre on both sides of the cut, and alignment confidence.
- **Mix compatibility** between two units, from any songs, is scored on: key (same, or within ±2 semitones of transposition), tempo (within ±8% stretch), meter, chord function (for example ii–V–I), swing ratio, mode, and which stems exist.

ID	The user can…	Done when	Priority	Status
M1	See the unit ladder on the timeline	Lanes for sections, phrases, bars, chord spans and beats, zoomable, with confidence shading	Must	Proposed
M2	Play or loop any unit	Clicking a unit loops it seamlessly, full mix or any stem	Must	Proposed
M3	Export a loop	Stems or mix cut on bar lines; tempo, key and chords in the metadata; true peak ⩽ −0.7 dBFS	Must	Proposed
M4	See and change a unit's mode	Written, embellished, improvised or comping shown per unit; changing it re-scores that unit only	Must	Proposed

ID	The user can…	Done when	Priority	Status
M5	See where the recording departs from the sheet	The route (repeats, extra choruses, cuts) shown against the sheet; low-confidence units flagged	Must	Proposed
M6	Search the library for units	Filter by level, key, tempo, chords, mode and stem, for example “4-bar ii–V–I in A minor, 140–160 bpm, improvised”	Should	Proposed
M7	Find units that fit a selected one	Ranked by compatibility; each auditions already matched in tempo and key	Should	Proposed
M8	Build a mix from units	Drop units on a timeline; tempo, key and swing match automatically; bounce at true peak ⩽ −0.7 dBFS	Should	Proposed

Done when

Batch alignment is FR-18 and loops and mixes are FR-19 in the PRD. Both are scored like every other deliverable.

- [ ] Every job stores a unit index from L1 to L10, with sheet reference, audio span, confidence and mode per unit
- [ ] Section order (the route) is correct on all three Lithuanian songs, including repeats and solo choruses
- [ ] Bar boundaries within 50 ms median, and chord spans ⩾ 85% correct, against musician-checked songs
- [ ] Note F-measure is computed only on written and embellished units; an improvised solo cannot change it
- [ ] One benchmark song per language is fully timed by a musician
- [ ] M1–M5 pass their walkthroughs; M2 loops play with no audible click on 20 random bar and phrase loops

Open questions

- [ ] Can the musicians who approved the three Lithuanian songs mark their form route and which spans are improvised? That is the fastest way to ground L8–L9 and the modes.
- [ ] Which of the three songs should be the fully timed Lithuanian benchmark?
- [ ] What is the smallest unit a user may export as a loop: a chord span, a bar, or a phrase?
- [ ] Mixing across songs: are tempo stretch (±8%) and transposition (±2 semitones) the right limits for this music?